

containerd v2.3β



An industry-standard container runtime

PromCon

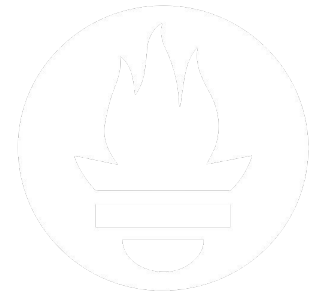
North America 2021

- CNCF Graduated since 2019
- Adopted by well-known enterprise Kubernetes services and distributions
- Designed to be pluggable to support cutting-edge features (remote snapshotters, WASM runtimes, ...)

Runtime plugins

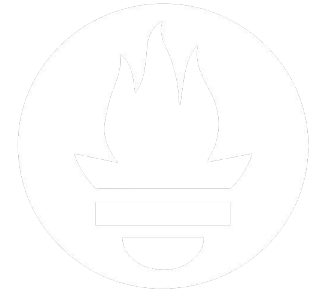
- Regular containers
 - runc: Linux
 - runhcs: Windows
 - runj: FreeBSD
- WebAssembly
 - runwasi
 - spin

- VM
 - Kata
 - nerdbox
 - supports macOS hosts
- Others
 - runsc (gvisor): ptrace sandbox, etc.
 - urunc: unikernel



PromCon
North America 2021

Snapshotter plugins



PromCon
North America 2021

Regular snapshotters: **overlayfs**, **btrfs**, **zfs**, **devmapper**, **erofs**, ...

“Remote” snapshotters support pulling image contents on demand to shorten the container startup time

- **stargz**, **soci**: Forward compatible with OCI v1 tar.gz images
- **nydus**: Uses an alternate image format
- **overlaybd**: Uses block devices as container images

nerdctl: contaiNERD CTL



- <https://github.com/containerd/nerdctl>
- Same UI/UX as the docker CLI (including Compose)
- Made for facilitating new experiments in the containerd platform (e.g., stargz, fast rootless with bypass4netns)
- Useful for debugging Kubernetes nodes too

```
$ nerdctl run hello-world  
$ nerdctl compose up
```

nerdbox: contaiNERD sandBOX



- <https://github.com/containerd/nerdbox>
- Linux containers on macOS
- microVM per container
- Based on libkrun
 - Network passthrough (TSI)
 - GPU passthrough (Venus)

	nerdbox	Lima
#VM : #container	1:1	1:N
containerd	Outside VM	Inside VM
Runtime	krun	runc
Filesystem	EROFS	OverlayFS

New release lifecycles

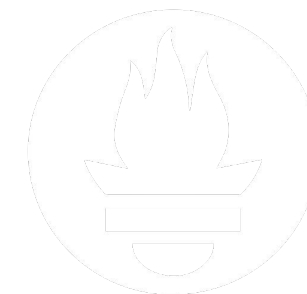
- **Regular:** every 4 months
 - Synced with Kubernetes
 - Support term: 8 months
- **Long-term support (LTS):** annually
 - Support term: 2 years

Apr 2026	v2.3 LTS
Aug 2026	v2.4
Dec 2026	v2.5
Apr 2027	v2.6 LTS

- The current v1.7 LTS (Mar 2023) will continue to be supported until Sep 2026



containerd v2.3 LTS (Apr 2026)

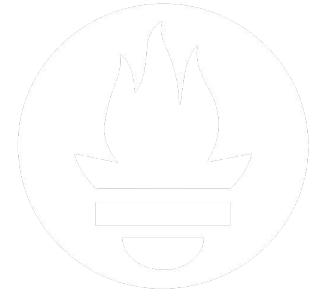


PromCon

North America 2021

- GPU vendor auto detection ([PR #12839](#))
- Support EROFS layers as an alternative to OCI tar layers ([PR #12567](#))
- More tight integration with SELinux in nerdctl ([nerdctl PR #4639](#), yet to be merged)
 - Support :z and :Z as in Docker and Podman
- And more! <https://github.com/containerd/containerd/releases/tag/v2.3.0-beta.0>

containerd v2.2 (Nov 2025)

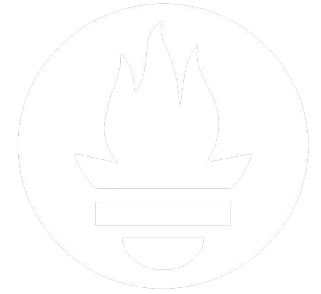


PromCon
North America 2021

- Mount manager
 - A mount for a container no longer need to be mountable on the host
 - Allows implementing EROFS snapshotter on macOS, without implementing EROFS driver for macOS
- WebAssembly plugins in Node Resource Interface (NRI)

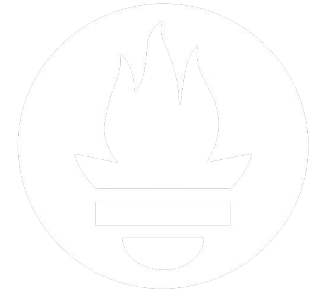
cgroup v1 Deprecation ! !

- Timeline: May 2029
- Am I affected? Run: `stat -fc %T /sys/fs/cgroup/`
 - If cgroupv2, the output is **cgroup2**
 - If cgroupv1, the output is **tmpfs**



PromCon
North America 2021

Join the community!



PromCon
North America 2021

- **GitHub:** <https://github.com/containerd>
- **Slack:** <https://slack.cncf.io/>
(Channels: `#containerd` `#containerd-dev`)
- **X (former Twitter):** [@containerd](https://twitter.com/containerd)
- **Bluesky:** [@containerd.dev](https://bsky.app/profile/containerd.dev)